

Lecture 17 Live Problems

Equilibria, generalized

Points toward lab manual pages: **GLV1, GLV2, GLV3, GLV4** (section L3 S5)

1. Four phase portraits. Find the equilibrium points and draw the phase portrait.

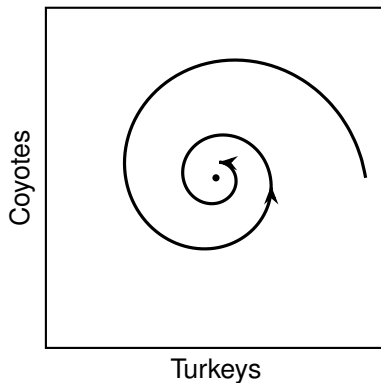
(A) $X' = 4 - X^2$

(B) $X' = 1 - X^2$

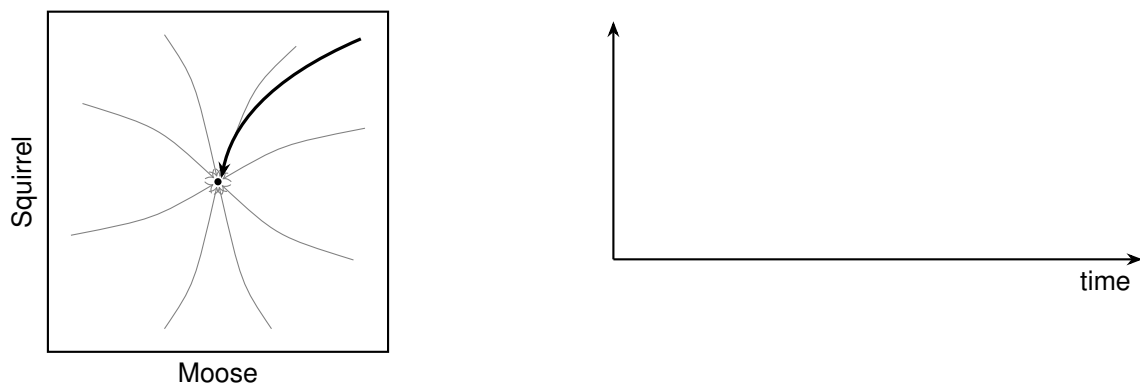
(C) $X' = 0 - X^2$

(D) $X' = -1 - X^2$

2. Turkeys and coyotes, spiraling in. Sketch the turkey and coyote time series associated to the following trajectory.



3. Moose and squirrel, straight in. Sketch the moose and squirrel time series associated to the following trajectory (the heavy curve).



4. Sharks and tuna with crowding. Draw the shark-nullcline and the tuna-nullcline, and find the equilibria of this system.

$$S' = -0.4S - 0.007S^2 + 0.03ST, \quad T' = 0.6T - 0.005T^2 - 0.04ST.$$

